

Student Name _____

CLO # 3: formulate and solve Constraint Satisfaction Problems in various domains, and constraints, and applying search techniques such as backtracking, checking, and arc consistency.

Q1. [15 marks] The organizing committee of an international Artificial Intelligence Conference is responsible for scheduling five major research presentations in a single-day program. Due to strict time management and speaker constraints, the committee must assign each presentation to a unique time slot while satisfying several logical and professional constraints.

The five presentations are: 1. Machine Learning (T1), 2. Robotics (T2), 3. Natural Language Processing (T3), 4. Computer Vision (T4), 5. Reinforcement Learning (T5)

The available time slots are:

Slot 1: 9:00 – 10:00 AM, Slot 2: 10:00 – 11:00 AM, Slot 3: 11:00 – 12:00 PM, Slot 4: 2:00 – 3:00 PM, Slot 5: 3:00 – 4:00 PM

Each presentation must be assigned exactly one slot, and no slot can host more than one presentation.

1. The Machine Learning (T1) and Natural Language Processing (T3) presentations must not be scheduled in consecutive time slots.
2. Robotics (T2) must be scheduled before Computer Vision (T4).
3. Reinforcement Learning (T5) cannot be scheduled in the morning slots (Slot 1 or Slot 2) due to speaker availability constraints.
4. Natural Language Processing (T3) must be scheduled after Machine Learning (T1).
5. Computer Vision (T4) cannot be placed in the final slot due to technical demonstration requirements.
6. Slot 3 is reserved for high-attendance presentations and can only be assigned to either Machine Learning (T1) or Robotics (T2).

Tasks:

- (a) Identify the variables of the CSP model. (4 Marks)
- (b) Define the domain for each variable. (4 Marks)

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- (c) Formally define all constraints (unary, binary, and ordering constraints). (8 Marks)
(d) Represent the constraint graph for the given problem. (4 Marks)
(e) Solve the CSP using backtracking and provide the final valid schedule. (5 Marks)

CL0 # 2: analyze adversarial environments and apply Min-Max and Alpha-Beta pruning algorithms to determine optimal strategies in competitive decision-making problems.

Q2. [15 marks] An AI agent is developed to play a strategic turn-based combat game against an opponent. In this environment, every decision leads to a possible battle outcome. The opponent is assumed to always play optimally to minimize the AI's advantage.

The AI agent uses the Minimax algorithm to determine the best possible strategy.

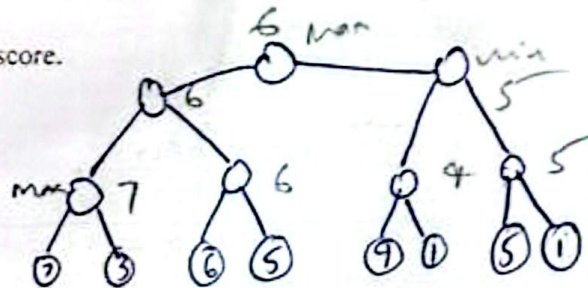
Instead of a traditional game tree, the game designer provides strategy cards, where each card represents a possible decision by the AI. Each strategy card has two possible opponent responses, each leading to different outcomes.

The AI acts as the MAX player, aiming to maximize its score.

The opponent acts as the MIN player, aiming to minimize the AI's score.

Strategy Cards (Game Outcomes):

- Card A:** Opponent responses $\rightarrow$ 7 or 3
Card B: Opponent responses $\rightarrow$ 6 or 5
Card C: Opponent responses $\rightarrow$ 9 or 1
Card D: Opponent responses $\rightarrow$ 5 or 11



Tasks:

- Explain how the given "strategy card" representation models a Minimax decision problem without using a tree structure. (3 Marks)
- For each card (A, B, C, and D), determine the optimal move of the opponent (MIN player) and justify your answer. (4 Marks)
- Based on the results obtained in part (b), determine which strategy the AI (MAX player) should select. (4 Marks)
- Identify the final guaranteed payoff (utility value) for the AI agent. (2 Marks)
- Explain why the Minimax algorithm assumes an optimal opponent and discuss one limitation of this assumption in real-world AI applications. (2 Marks)

CLO # 3: EDA, preprocessing techniques and apply supervised and unsupervised machine learning algorithms and evaluate model performance using standard metrics and validation techniques.

Q3. [20 marks] A retail company has provided transactional-level sales data containing invoice details, product information, and customer attributes. The dataset includes variables such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. The objective of this task is to perform advanced exploratory data analysis (EDA), clean and transform the dataset, and prepare meaningful features that can later be used for machine learning models such as clustering or predictive analytics.

Part 1 EDA and Preprocessing

1. Load the dataset and display structure (shape, datatypes, summary statistics). (1 Marks)
2. Identify and handle missing values in CustomerID and Description using appropriate strategies (1 Marks)
3. Remove invalid transactions such as negative Quantity or zero/negative UnitPrice. (1 Marks)
4. Convert InvoiceDate into datetime format and extract new features (month, weekday, hour). (1 Marks)
5. Create a new feature TotalPrice = Quantity × UnitPrice. (1 Marks)

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6. Detect and treat outliers in Quantity and UnitPrice using IQR or Z-score method. (1 Marks) ?
7. Analyze sales distribution across different countries using visualization. (1 Marks)
8. Identify top-selling products using StockCode and visualize results. (1 Marks)?
9. Perform time-based analysis to find peak sales periods (daily or monthly trend). (1 Marks)?
10. Display the number of transactions per customer or per invoice to understand customer purchasing behavior. (1 Marks) ?

Part 2 Model Training and Evaluation

1. Prepare a customer-level dataset by creating aggregated features such as total spend, purchase frequency, and total quantity per customer. (1 Marks)
2. Split the dataset into training and testing sets using an appropriate ratio. (1 Marks)
3. Apply feature scaling (standardization) to numerical features before clustering. (1 Marks)
4. Apply K-Means clustering on the training dataset and fit the model. (1 Marks)
5. Determine the optimal number of clusters using the Elbow Method and select the best value. (1 Marks)
6. Assign cluster labels to the training data and store cluster centroids from the trained model. (1 Marks)
7. Predict cluster labels for the test dataset using the trained K-Means model. (1 Marks)
8. Compare and analyze cluster assignment consistency between training and testing datasets. (1 Marks)
9. Visualize clusters for both training and testing datasets using PCA method and interpret the results. (1 Marks)